

Group C Practical 1

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import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin

URL = "https://www.goodreads.com/list/show/264.Books_to_Read"
HEADERS = {
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/117.0.0.0 Safari/537.36'
}

def fetch_page(url):
    """Fetches the HTML of a page and returns the response text.
    Raises requests.RequestException on network errors.
    """
    resp = requests.get(url, headers=HEADERS, timeout=10)
    resp.raise_for_status()
    return resp.text

def parse_books(html, base_url):
    """Parses book titles and relative links from the Goodreads list HTML.

    Returns a list of tuples: (title, absolute_url)
    """
    soup = BeautifulSoup(html, 'html.parser')

    # Goodreads list pages contain book titles inside <a class="bookTitle" ...>
    # We'll also fall back to a <span class="bookTitle"> if present (older markup)
    books = []

    # Primary selector: anchor elements with class bookTitle
    for a in soup.select('a.bookTitle'):
        title = a.get_text(strip=True)
        href = a.get('href')
        if not href:
            continue
        full = urljoin(base_url, href)
        books.append((title, full))

    # If none found, try spans with bookTitle (less common)
    if not books:
        for span in soup.select('span.bookTitle'):
            title = span.get_text(strip=True)
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        parent_a = span.find_parent('a')
        href = parent_a.get('href') if parent_a else None
        if not href:
            continue
        full = urljoin(base_url, href)
        books.append((title, full))

    return books

def main():
    try:
        html = fetch_page(URL)
    except requests.RequestException as e:
        print(f"Network error fetching {URL}: {e}")
    return

    books = parse_books(html, URL)
    if not books:
        print("No books found on the page. The page structure may have changed or content is blocked.")
    return

    for i, (title, link) in enumerate(books, start=1):
        print(f"{i}. {title} - {link}")

if __name__ == '__main__':
    main()

```

Output

1. To Kill a Mockingbird - https://www.goodreads.com/book/show/2657.To_Kill_a_Mockingbird
2. Pride and Prejudice - https://www.goodreads.com/book/show/1885.Pride_and_Prejudice
3. The Diary of a Young Girl - https://www.goodreads.com/book/show/48855.The_Diary_of_a_Young_Girl
4. 1984 - <https://www.goodreads.com/book/show/61439040-1984>
5. Animal Farm - https://www.goodreads.com/book/show/170448.Animal_Farm
6. Harry Potter and the Philosopher's Stone (Harry Potter, #1) - <https://www.goodreads.com/book/show/42844155-harry-potter-and-the-philosopher-s-stone>
7. The Little Prince - https://www.goodreads.com/book/show/157993.The_Little_Prince
8. The Great Gatsby - https://www.goodreads.com/book/show/4671.The_Great_Gatsby

9. The Catcher in the Rye -
https://www.goodreads.com/book/show/5107.The_Catcher_in_the_Rye
10. The Lord of the Rings -
https://www.goodreads.com/book/show/33.The_Lord_of_the_Rings
11. The Book Thief - https://www.goodreads.com/book/show/19063.The_Book_Thief
12. Jane Eyre - https://www.goodreads.com/book/show/10210.Jane_Eyre
13. Harry Potter and the Sorcerer's Stone (Harry Potter, #1) -
https://www.goodreads.com/book/show/3.Harry_Potter_and_the_Sorcerer_s_Stone
14. The Chronicles of Narnia (The Chronicles of Narnia, #1-7) -
https://www.goodreads.com/book/show/11127.The_Chronicles_of_Narnia
15. Lord of the Flies - https://www.goodreads.com/book/show/7624.Lord_of_the_Flies
16. Romeo and Juliet - https://www.goodreads.com/book/show/18135.Romeo_and_Juliet
17. Harry Potter and the Deathly Hallows (Harry Potter, #7) -
https://www.goodreads.com/book/show/136251.Harry_Potter_and_the_Deathly_Hallows
18. The Kite Runner - https://www.goodreads.com/book/show/77203.The_Kite_Runner
19. The Giver (The Giver, #1) - https://www.goodreads.com/book/show/3636.The_Giver
20. The Giving Tree - https://www.goodreads.com/book/show/370493.The_Giving_Tree
21. The Hunger Games (The Hunger Games, #1) -
<https://www.goodreads.com/book/show/2767052-the-hunger-games>
22. Little Women (Little Women, #1) -
https://www.goodreads.com/book/show/1934.Little_Women
23. Charlotte's Web - https://www.goodreads.com/book/show/24178.Charlotte_s_Web
24. The Hobbit, or There and Back Again -
https://www.goodreads.com/book/show/5907.The_Hobbit_or_There_and_Back_Again
25. Fahrenheit 451 - <https://www.goodreads.com/book/show/13079982-fahrenheit-451>
26. Of Mice and Men - https://www.goodreads.com/book/show/890.Of_Mice_and_Men
27. Harry Potter and the Prisoner of Azkaban (Harry Potter, #3) -
https://www.goodreads.com/book/show/5.Harry_Potter_and_the_Prisoner_of_Azkaban
28. Green Eggs and Ham -
https://www.goodreads.com/book/show/23772.Green_Eggs_and_Ham
29. Wuthering Heights - https://www.goodreads.com/book/show/6185.Wuthering_Heights
30. The Picture of Dorian Gray -
https://www.goodreads.com/book/show/5297.The_Picture_of_Dorian_Gray
31. Alice's Adventures in Wonderland / Through the Looking-Glass -
https://www.goodreads.com/book/show/24213.Alice_s_Adventures_in_Wonderland_Through_the_Looking_Glass
32. Night - <https://www.goodreads.com/book/show/1617.Night>
33. Gone with the Wind -
https://www.goodreads.com/book/show/18405.Gone_with_the_Wind
34. The Holy Bible: King James Version -
https://www.goodreads.com/book/show/1923820.The_Holy_Bible
35. Harry Potter and the Goblet of Fire (Harry Potter, #4) -
https://www.goodreads.com/book/show/6.Harry_Potter_and_the_Goblet_of_Fire

36. The Hitchhiker's Guide to the Galaxy (The Hitchhiker's Guide to the Galaxy, #1) -
https://www.goodreads.com/book/show/386162.The_Hitchhiker_s_Guide_to_the_Galaxy
37. Hamlet - <https://www.goodreads.com/book/show/1420.Hamlet>
38. Harry Potter and the Half-Blood Prince (Harry Potter, #6) -
<https://www.goodreads.com/book/show/58613345-harry-potter-and-the-half-blood-prince>
39. The Adventures of Huckleberry Finn -
https://www.goodreads.com/book/show/2956.The_Adventures_of_Huckleberry_Finn
40. Tolkien On Fairy-stories -
https://www.goodreads.com/book/show/1362112.Tolkien_On_Fairy_stories
41. Les Misérables - https://www.goodreads.com/book/show/24280.Les_Mis_rables
42. Brave New World and Brave New World Revisited -
https://www.goodreads.com/book/show/5479.Brave_New_World_and_Brave_New_World_Revisited
43. Crime and Punishment -
https://www.goodreads.com/book/show/7144.Crime_and_Punishment
44. The Help - <https://www.goodreads.com/book/show/4667024-the-help>
45. The Secret Garden - https://www.goodreads.com/book/show/2998.The_Secret_Garden
46. The Lion, the Witch and the Wardrobe (Chronicles of Narnia, #1) -
https://www.goodreads.com/book/show/100915.The_Lion_the_Witch_and_the_Wardrobe
47. Memoirs of a Geisha -
https://www.goodreads.com/book/show/929.Memoirs_of_a_Geisha
48. The Odyssey - https://www.goodreads.com/book/show/1381.The_Odyssey
49. A Christmas Carol - https://www.goodreads.com/book/show/5326.A_Christmas_Carol
50. The Cat in the Hat (Cat in the Hat, #1) -
https://www.goodreads.com/book/show/233093.The_Cat_in_the_Hat
51. The Outsiders - https://www.goodreads.com/book/show/231804.The_Outsiders
52. One Hundred Years of Solitude -
https://www.goodreads.com/book/show/320.One_Hundred_Years_of_Solitude
53. A Thousand Splendid Suns -
https://www.goodreads.com/book/show/128029.A_Thousand_Splendid_Suns
54. Anne of Green Gables (Anne of Green Gables, #1) -
https://www.goodreads.com/book/show/8127.Anne_of_Green_Gables
55. The Grapes of Wrath - <https://www.goodreads.com/book/show/18114322-the-grapes-of-wrath>
56. The Elephant Tree - <https://www.goodreads.com/book/show/9293020-the-elephant-tree>
57. The Count of Monte Cristo -
https://www.goodreads.com/book/show/7126.The_Count_of_Monte_Cristo
58. Where the Sidewalk Ends -
https://www.goodreads.com/book/show/30119.Where_the_Sidewalk_Ends
59. Ender's Game (Ender's Saga, #1) -
https://www.goodreads.com/book/show/375802.Ender_s_Game

60. Where the Wild Things Are -
https://www.goodreads.com/book/show/19543.Where_the_Wild_Things_Are
61. The Color Purple - <https://www.goodreads.com/book/show/52892857-the-color-purple>
62. The Handmaid's Tale (The Handmaid's Tale, #1) -
https://www.goodreads.com/book/show/38447.The_Handmaid_s_Tale
63. The Adventures of Tom Sawyer -
https://www.goodreads.com/book/show/24583.The_Adventures_of_Tom_Sawyer
64. A Tale of Two Cities -
https://www.goodreads.com/book/show/1953.A_Tale_of_Two_Cities
65. Life of Pi - https://www.goodreads.com/book/show/4214.Life_of_Pi
66. Frankenstein: The 1818 Text - <https://www.goodreads.com/book/show/35031085-frankenstein>
67. One Flew Over the Cuckoo's Nest -
https://www.goodreads.com/book/show/332613.One_Flew_Over_the_Cuckoo_s_Nest
68. Brave New World - https://www.goodreads.com/book/show/5129.Brave_New_World
69. Anna Karenina - <https://www.goodreads.com/book/show/15823480-anna-karenina>
70. Oh, the Places You'll Go! -
https://www.goodreads.com/book/show/191139.Oh_the_Places_You_ll_Go_
71. Tuesdays with Morrie: An Old Man, a Young Man, and Life's Greatest Lesson -
https://www.goodreads.com/book/show/6900.Tuesdays_with_Morrie
72. The Fellowship of the Ring (The Lord of the Rings, #1) -
<https://www.goodreads.com/book/show/61215351-the-fellowship-of-the-ring>
73. The Zombie Room - <https://www.goodreads.com/book/show/29908754-the-zombie-room>
74. Slaughterhouse-Five -
https://www.goodreads.com/book/show/4981.Slaughterhouse_Five
75. The Old Man and the Sea -
https://www.goodreads.com/book/show/2165.The_Old_Man_and_the_Sea
76. Winnie-the-Pooh (Winnie-the-Pooh, #1) -
https://www.goodreads.com/book/show/99107.Winnie_the_Pooh
77. Catch-22 - https://www.goodreads.com/book/show/168668.Catch_22
78. The Girl With the Dragon Tattoo (Millennium, #1) -
https://www.goodreads.com/book/show/2429135.The_Girl_With_the_Dragon_Tattoo
79. The Scarlet Letter - https://www.goodreads.com/book/show/12296.The_Scarlet_Letter
80. The Alchemist - https://www.goodreads.com/book/show/865.The_Alchemist
81. Macbeth - <https://www.goodreads.com/book/show/8852.Macbeth>
82. The Stranger - https://www.goodreads.com/book/show/49552.The_Stranger
83. Twilight (The Twilight Saga, #1) -
<https://www.goodreads.com/book/show/41865.Twilight>
84. The Metamorphosis -
https://www.goodreads.com/book/show/485894.The_Metamorphosis
85. Lolita - <https://www.goodreads.com/book/show/7604.Lolita>
86. Siddhartha - <https://www.goodreads.com/book/show/52036.Siddhartha>

87. The Time Traveler's Wife - <https://www.goodreads.com/book/show/18619684-the-time-traveler-s-wife>
88. The Girl Who Played with Fire (Millennium #2) - <https://www.goodreads.com/book/show/5060378-the-girl-who-played-with-fire>
89. Dark Places - <https://www.goodreads.com/book/show/5886881-dark-places>
90. Gone Girl - <https://www.goodreads.com/book/show/19288043-gone-girl>
91. Goodnight Moon - https://www.goodreads.com/book/show/32929.Goodnight_Moon
92. A Story of Yesterday - <https://www.goodreads.com/book/show/58913358-a-story-of-yesterday>
93. Harry Potter and the Chamber of Secrets (Harry Potter, #2) - https://www.goodreads.com/book/show/15881.Harry_Potter_and_the_Chamber_of_Secrets
94. The Girl on the Train - <https://www.goodreads.com/book/show/22557272-the-girl-on-the-train>
95. A Clockwork Orange - <https://www.goodreads.com/book/show/41817486-a-clockwork-orange>
96. War and Peace - https://www.goodreads.com/book/show/656.War_and_Peace
97. Shutter Island - https://www.goodreads.com/book/show/21686.Shutter_Island
98. A Tree Grows in Brooklyn - https://www.goodreads.com/book/show/14891.A_Tree_Grows_in_Brooklyn
99. Mystic River - https://www.goodreads.com/book/show/21671.Mystic_River
100. Lock Every Door - <https://www.goodreads.com/book/show/41837243-lock-every-door>

=== Code Execution Successful ===